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Venture Capitalists' Assessment of New Venture Survival

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This study investigates whether VCs' assessment policies of new venture survival are consistent with those arising from the strategy literature (using two established strategy perspectives). Strategy scholars suggest the nature of the markets, competition, and decisions made by the management team affect a new venture's survival chances. The findings demonstrate that VCs' assessment policies are predominantly consistent with those proposed by strategy scholars—providing insight into why VCs consider certain criteria in their assessment of new venture survival as well as why some criteria are more important in their assessment than others. Through this increased understanding of venture capitalists' decision making, entrepreneurs seeking capital may be better able to address their requests for funding to those criteria venture capitalists find most critical to the survival of a new venture. Venture capitalists may use these findings to better understand their own decision making process, which, in turn, provides the opportunity to increase evaluation efficiency.

(Venture Capital; Decision Models; New Venture Strategy; Survival; Conjoint Analysis)

Introduction

The majority of research on VCs' decision making has produced empirically derived lists of VCs' "espoused" criteria—criteria they report they use when evaluating either global measures of performance (c.f., MacMillan et al. 1987) or likely profitability (c.f., Roure and Keeley 1990). Few researchers have investigated VCs' assessment of new venture survival. This is surprising considering Tyebjee and Bruno's (1984) finding that VCs' investment decisions can be predicted from their perceptions of risk (new venture failure) and return. It appears that global measures of new venture performance may fail to capture the complexity of a VC's task. On the other hand, although studies investigating VCs' assessments of profitability are no doubt advancing the field, they need to be supplemented with studies of probability of survival assessments.

There is substantial evidence from strategy research that suggests strategic factors impact the survival

chances of new ventures (c.f., Lieberman and Montgomery 1988, Bruderl et al. 1992). However, do VCs follow the decision logic reported in strategy research? The theory from the strategy literature used in this study is contingent upon the issues raised by the "industrial organization" (IO) and "population ecology" perspectives. The IO strategy perspective deals with the competitive positioning of the firm. The population ecology perspective deals with the firm's liabilities of newness—encapsulating those factors that have the greatest impact on new firm mortality. The concomitant consideration of these issues enables us to determine whether the VCs' assessment of a venture is consistent with whether the venture will actually survive (as predicted by strategy research). This approach addresses some of the theoretical limitations of previous research into VCs' decision making.

However, limitations of previous research into VC

decision making also arise from the research methods used. Such limitations include problems of retrospective reporting (c.f., Tyebjee and Bruno 1984), use of questionnaire responses rather than actual evaluations (c.f., MacMillan et al. 1987) and biases and errors associated with self-reporting (c.f., Hisrich and Jankowitz 1990). Previous research has typically investigated what Argyris and Schon (1974) refer to as "espoused theories of action" as opposed to "theories in use."

Furthermore, few researchers to date have acknowledged that there may be a difference in a VC's decision policy for businesses in different stages of development. For example, we can expect that a VC's assessment policy for a new venture searching for seed, start up, or development capital may be different from the assessment policy for an established business searching for mezzanine capital.

This study addresses some of the theory and research method limitations of previous research into VC decision making by investigating whether VCs' "in use" assessment policies of new venture survival are consistent with those arising from the strategy literature predominantly from an IO strategy perspective.

The study proceeds as follows: First, the strategy literature is reviewed and hypotheses are generated and directed at the research question above. Second, data collection is discussed, including a description of the sample and the survey instrument used. Third, conjoint analysis is used to determine if the strategic considerations suggested by strategy scholars are actually those used by VCs to predict a venture's survival. Fourth, the results are reviewed and discussed in terms of the strategy literature and our understanding of VCs' decision making. Finally, the implications of this study to future research and to VCs and entrepreneurs are discussed.

Literature Review: Strategic Management

There is substantial evidence from the strategy literature that strategic factors impact the performance/survival chances of new ventures (c.f., Lieberman and Montgomery 1988). Central to IO strategy research is an understanding of the nature of the **markets** (Williamson 1985,

Golder and Tellis 1993) in which a new venture will compete and the nature of **competitive** interactions between firms in those markets (Schmalensee 1982, Yip 1982). The **management** team must then decide where to position the new venture in order to achieve its competitive advantage (Besanko et al. 1996). In conjunction with this IO strategy perspective, we consider findings from a population ecology perspective of strategy that as new ventures learn over time about their internal and external environments they increase their probability of survival.

The literature review below addresses considerations of the market, considerations of competition, and finally the ability of management to choose and establish a viable position. A series of hypotheses are generated that propose VCs follow the decision logic proposed by strategy scholars. These hypotheses are then empirically investigated.

Market Considerations

IO strategy researchers propose that superior performance arises from a fit between the competencies of a venture and the key success factors of an industry (Andrews 1987). **Key success factors** represent the requirements necessary for success within a particular industry. Therefore a new venture team must commit to a number of key factors they believe will lead to success within the competitive environment (Slater 1993). However, if the competitive environment changes, so too may the key success factors rendering the venture at a competitive disadvantage (Aaker and Day 1986, Golder and Tellis 1993). For example, if a new venture commits to a new technology, it faces the risk that as alternate technologies emerge, its technology will not perform as expected, and/or there will be little or no demand (Williamson 1985, Aaker and Day 1986). Similarly, shifts in consumers' tastes may place the new venture at a competitive disadvantage (Golder and Tellis 1993). Therefore if the key success factors are **unstable** (i.e., requirements necessary for success will likely change radically during industry development), a new venture is likely to face a high degree of technical and demand uncertainty (Nelson and Winter 1982).

If firms could readily adapt to changing key success factors, a "good fit" with the environment could be

maintained. However, organizational inertia makes such change difficult (Boeker 1989). For example, there is a tendency to counter environmental threats by investing research funds into improving the current technology (Aaker and Day 1986), and / or solving new problems using inappropriate methods (Nelson and Winter 1982). These responses are often combined with a reluctance to withdraw too quickly from mature technologies that are highly profitable (Yip 1982).¹

On the other hand, if key success factors within an industry remain **stable** (i.e., *requirements necessary for success are unlikely to change radically during industry development*), early commitment to a new technology and customers is likely to provide a sustainable competitive position and thereby a higher probability of survival. **Survival** is defined as: *the probability that this venture will continue to participate in the market*. The above discussion leads to Hypothesis 1.

*Hypothesis 1. VCs' assessment of the probability of survival is significantly higher for ventures facing stable key success factors than those facing unstable key success factors.*²

Common wisdom from the strategy literature suggests that timing of entry, as well as key success factor stability, affects performance—pioneers have higher returns if they are successful (DeCastro and Chrisman 1995), but bear a higher risk of failure (Aaker and Day 1986, Mitchell 1991). A **pioneer** enters a new industry first and a **late follower** enters an industry late in the industry's stage of development. The higher risk of failure for pioneers, relative to later followers, is associated with higher levels of uncertainty. For instance, as a pioneer creates an industry, there is often uncertainty whether there will be sufficient market to sustain survival. Furthermore, customers may be reluctant to substitute out of a market in which they feel comfortable and into one whose future is uncertain (Lambkin and Day 1989).

¹ Decisions to persist with outdated technology are often poor investments due to the law of diminishing returns (Foster 1982) and difficulty associated with transferring the link with old customers to new opportunities (Mitchell 1991).

² The levels and definitions for all attributes are displayed in Appendix A.

Late followers are better able to recognize the attractiveness of a market and are able to minimize the costs of entry through cutting R&D corners and / or leap-frogging the pioneering technology (Yip 1982, Harrison and Fiet forthcoming). Some entrepreneurs decide not to pioneer with unproven technology or an unproven market, but wait until more information is available about customers' tastes and behavior so as to provide a product that better meets their needs (Peters and Waterman 1982). In doing so they learn from the pioneer's mistakes (Prahalad and Hamel 1990). This increased certainty, coupled with a reduced strain on resources, increases the probability of survival of later entrants over pioneers (Mitchell 1991).³ The above discussion leads to Hypothesis 2:

Hypothesis 2. VCs' assessment of the probability of survival is higher for later followers than for pioneers.

Competition Considerations

In newly established industries, pioneers need to create and develop entry barriers to avoid having their positions eroded by new competitors who can successfully imitate them. Entry barriers provide the pioneer an opportunity to operate in their industry for a grace period under conditions of limited competitive rivalry (with the possible exception of substitute products). Therefore barriers to entry initially provide a pioneer a lead time, and thereafter minimize competitive rivalry within the industry. **Lead time** refers to the period of monopoly for the first entrant prior to competitors entering the industry and **competitive rivalry** refers to the level of competition among industry members during industry development.

Together, lead time and subsequent competitive rivalry provide greater understanding of new venture performance by identifying how an advantage is obtained and the means by which it slowly reduces over time.⁴ High barriers to entry can lengthen a new venture's lead time, providing important advantages

³ Of course later entry may restrict the venture's upside potential (profitability), which is beyond the scope of this paper.

⁴ Barriers to entry provide a competitive shield (or act as an isolating mechanism) for industry members and may take many forms; for example, technological leadership (Lieberman and Montgomery 1988); consumer-based information advantages (Schmalensee

including helping the venture strengthen its brand name (Schmalensee 1982), broaden its product line (Robinson and Fornell 1985), and achieve cost advantages through experience effects (Abell and Hammond 1979).

A long lead time (i.e., *an extended period of monopoly*) also provides time for a venture to learn new tasks, to invent and overcome conflict in new roles, to develop an informal structure, to create stable links with stakeholders, and to develop some organizational inertia and stability that will encourage customer trust. In other words, in the absence of industry competitors, lead time allows a new venture to increase its probability of survival (Freeman et al. 1983, Carroll and Huo 1986, Halliday et al. 1987). The subsequent level of competitive rivalry can also impact performance—as the level of competitive rivalry increases, so too does the risk of business failure (Keeley et al. 1996). This discussion is reflected in the following two hypotheses.

Hypothesis 3. *VCs' assessment of the probability of survival is higher for long lead times than for short lead times.*

Hypothesis 4. *VCs' assessment of the probability of survival is significantly higher for ventures facing low competitive rivalry than for those facing high competitive rivalry.*

Management Capability Considerations

As discussed under market considerations there is often considerable uncertainty about future demand (Fiet 1995). New ventures often need to educate the market about their product and persuade potential customers that the benefits of purchase are greater than the risks (Slater 1993). Once a frame of reference has been created and customers are convinced that product benefits outweigh the risks associated with newness, the new venture may have been able to establish a market presence and the legitimacy that comes with it.

However, a customer's frame of reference and market legitimacy can be difficult and costly (in terms of

both human and financial resources) for the new venture to construct. If a venture already possesses these resources, it has educational capability that can be directed towards performing original market research and necessary market development (Stinchcombe 1965). Therefore **educational capability** refers to the amount of resources and skills available to overcome market ignorance through education.

Ventures with high educational capability can more rapidly develop stable links with stakeholders, decrease customer uncertainty, and engender customer trust, thereby increasing the firm's probability of survival (Stinchcombe 1965, Hannan and Freeman 1989). Furthermore, a strong effort at educating the market may also signal to customers the commitment of the new venture's managers to their product (Bradenburger and Nalebluff 1996). The above discussion leads to Hypothesis 5.

Hypothesis 5. *VCs' assessment of the probability of survival is significantly higher for ventures with high educational capability than for those with low educational capability.*

Although VCs are expected to consider management's educational capability in their assessment of likely new venture survival, Dunne et al. (1989) propose that firms also need to be investigated in light of management experience in related industries. **Industry related competence** refers to the level of experience and knowledge with the industry being entered, or a related industry. It appears success is more likely to be achieved by those entering an industry in which the management team has prior experience (Roure and Maidique 1986, Cooper 1986). Roure and Maidique (1986) found that successful founders had experience in rapid growth firms that competed in the same industry as the start up. Opportunities often occur too quickly to be able to be grasped by someone from outside the industry; they must possess the necessary skills a priori (Feesser and Willard 1990). It appears industry specific human capital is a significant determinant of venture survival (Cooper et al. 1988, Bruderl et al. 1992). The above discussion leads to Hypothesis 6.

Hypothesis 6. *VCs' assessment of the probability of survival is significantly higher for ventures with high*

1982); and exclusive access to certain distribution channels (Karakaya and Kobu 1994).

industry related competence than for those with low industry related competence.

Relative Importance of Decision Criteria

The current literature on VCs' decision making suggests the management team is important and often ranked as the most important criterion (c.f., Dixon 1991, MacMillan et al. 1987). Why is the quality of the management team so important to VCs? The strategy literature above suggests that uncertainty plays an important role in the survival chances of new ventures; for example, uncertainty over demand, technology, and competition as well as customers' uncertainty about a new venture's market offerings. We propose that while the market considerations (key success factor stability and timing) and competition considerations (lead time and competitive rivalry) are no doubt important, especially in retrospect, they are difficult for a VC to assess in prospect. They are difficult to predict due to the enormous environmental uncertainty typically surrounding a new venture. We propose one way a VC manages such uncertainty is to choose a management team they believe will be able to cope with expected and unexpected changes in the market and competitive environment, i.e., a management team with high educational capability and high industry related competence. This leads to the final hypothesis:

Hypothesis 7(a). *In their assessment of new venture survival, VCs place higher importance on industry related competence than they do on key success factor stability, timing of entry, lead time, or competitive rivalry.*

Hypothesis 7(b). *In their assessment of new venture survival, VCs place higher importance on educational capability than they do on key success factor stability, timing of entry, lead time, or competitive rivalry.*

In sum, strategy researchers (utilizing primarily an IO strategy perspective) suggest that the nature of the market, the competitive interactions between firms, and the abilities of the management team to establish a viable position in the market are central to understanding firm survival. This leads to a number of hypothesized relationships that we would expect VCs to consider if their decision policies are consistent with

whether the venture actually will survive (as predicted by strategy research). Specifically, VCs' assessment of probability of survival is significantly higher for ventures facing high key success factor stability, that are late followers, have long lead times, low competitive rivalry, high educational capability, and high industry related competence. The research method to empirically test the above hypothesized relationships is now described.

Research Method

Sampling Plan, Survey Method, and Sample. The 1996 Australian Venture Capital Guide (Pollitecon Publications 1996), a comprehensive guide to Australia's venture capital industry, and the 1995 Australian Development Capital Directory (Australian Department of Industry, Science and Technology 1995) were used to identify potential firms and participants. Firms listed in either guide and based in Australia with preferred investments in seed, start up, and/or development capital were considered as the sampling frame of VCs in Australia. The senior executives of each firm in the sampling plan were targeted for this study. It is estimated the Australian venture capital industry consists of 65 private venture capital firms with a total of 350 professional staff, \$4 billion invested capital, \$1 billion available capital, a current portfolio of 958 companies, with 655 completed investments (Pollitecon Publications 1996).⁵

Two methods of data collection were used: (1) experiment collected by author, and (2) experiment sent by mail. For reasons of resource efficiency the author personally collected only those responses from VCs located in Australia's three largest cities. Analysis of variance was performed on the importance weights of those responses collected through mail and those collected by the author. The two methods of data collection were not significantly different, consequently the two groups of responses were treated as one.

⁵ There are also six government or semigovernment institutions involved in the equity funding of new ventures. These six firms were also included in the sampling plan. There was no significant difference in the decision making of private and public VCs, although small sample sizes for this statistical test would make it difficult to detect such a difference.

Sixty-six individual VCs representing 47 venture capital firms completed the survey capturing Australia's most senior VCs with Directors, Managing Directors, Executive Chairmen, General Managers, and CEOs representing 52% of the sample. Senior managers constitute a further 42% with only 6% of the sample accounted for by analysts. This represents a 65% response rate in terms of participating venture capital firms and a 19% response rate in terms of total professional VCs.⁶ A comparison of VC firms in the study and those that did not participate reveal no significant difference in year established, number of professional staff, types of investment, total capital, completed divestments, and current number of portfolio companies (comparison data obtained from the Australian Venture Capital Guide (Pollitecon Publications 1996)). Given the similarity between participating and nonparticipating firms there is unlikely to be a nonresponse bias.

Attributes, Levels, and Dependent Variable. In this study, VCs evaluate a series of hypothetical conjoint profiles which describe new ventures in terms of eight attributes, each with two levels: (1) Stability of Key Success Factors, (2) Timing of Entry, (3) Lead Time, (4) Competitive Rivalry, (5) Industry Related Competence, (6) Educational Capability, (7) Mimicry, and (8) Scope. The attributes of mimicry and scope were used as control variables. Attributes, levels, and definitions are provided earlier in the literature review. Attribute levels were chosen to represent variation that typically occurs in the decision environment of VCs, thereby maintaining believability and response validity. A pretest with VCs, accountants, and academics confirmed the face validity for both the attributes and their levels.

Survival (the probability that this venture will continue to participate in the market using a ten-year time horizon) is represented by an eleven-point scale with end anchors describing "very low probability of survival" and "very high probability of survival." Con-

tinued market participation is frequently used as a definition of survival (Lambkin and Day 1989). Biggadike (1979) and McDougall et al. (1994) find that, after eight years, new ventures begin to resemble the characteristics of an established venture. A ten-year time horizon was chosen, as it appears reasonable to assume ten years captures the transition from a new venture to an established business.⁷

Research Instrument and Experimental Design.

The research instrument contained a cover letter, task instructions, the conjoint decision making experiment, and a post-experiment questionnaire that asked respondents to answer questions regarding characteristics of themselves and their firm. Relevant term definitions were also included on a detachable sheet that could be referred to while completing the survey. Once instructions were understood, respondents considered each conjoint venture description and provided a rating on an 11-point scale for the dependent measure—probability of survival.

For the conjoint experiment, an orthogonal fractional factorial design was used to reduce the number of attribute combinations and thus make the decision making task more manageable (Green and Srinivasan 1990). Each of the eight attributes was varied at two levels in a fractional factorial design consisting of 16 profiles (Hahn and Shapiro 1966). The original profiles were replicated to permit estimates of individual subject error for use in subsequent statistical analysis and allow a test-retest measure of reliability. These 32 profiles were randomly assigned to avoid order effects. A practice case and six hold-out cases were also used. The practice case familiarized respondents with the task and the six hold-out cases were used to test the models' predictive ability. Therefore the experiment presented VCs with 39 profiles to evaluate. The experimental design enabled both individual subject level and aggregate subject-level analyses.

Analysis. This study uses conjoint analysis to determine if strategic VC considerations reported by strategy scholars are actually those used to predict a

⁶ Some firms had more than one VC participate in the study. The results of this study did not vary significantly when the data were reanalyzed including the responses of only one VC per venture capital firm.

⁷ Although VCs typically harvest the business within eight years, they are often selling future cash flows extending to, and beyond, this ten-year period.

new venture's survival. **Conjoint analysis** is a general term referring to a technique that requires respondents to make a series of judgments based on a set of attributes (cues) from which the underlying structure of their cognitive system can be investigated (Shepherd and Zacharakis 1997). From this series of judgments, the respondent's decisions can be decomposed, thus providing the researcher an opportunity to investigate the underlying structure of the decisions. Importantly, this technique avoids the use of retrospective and self-reported data by collecting information about a decision as that decision is made (Zacharakis and Meyer 1998).

Regression and analysis of variance (ANOVA) are the statistical techniques used to decompose the decision, e.g., regression decomposes an assessment into its underlying structure as represented by the independent variables and their corresponding beta coefficients. The conjoint technique allows analysis at both the individual and aggregate subject level, which improves the predictive ability of the research (c.f., Moore 1980).

To identify attributes statistically significant at the aggregate level, the regression coefficient (β) for each attribute (derived from the individual-subject level of analysis) are averaged across individuals with the sign of the β indicating the nature of the relationship (Vancouver and Morrison 1995). The mean β s represent a model of the sample's decision making. A **Z-statistic** aggregates the *t*-statistics derived from the individual-subject analysis for that attribute to identify whether a particular attribute is significantly used by the sample (Dechow et al. 1994).

Although two or more attributes may significantly affect the decision process, it is unlikely that those attributes will be of equal importance. Therefore the significance at the aggregate level of analysis is supplemented with a measure of relative importance—Hays' (1973) omega squared (ω^2). **Omega Squared** is a measure of explained variance, and is used to assess the relative importance of the attributes. The mean omega-squared values corresponding to all main effects were calculated. Predictive ability of individual and aggregate decision making models were tested using a Pearson *R* correlation between the observed score on six hold-out profiles and a predicted score calculated

by the decision model(s). A VC's decision model contained a constant and a regression coefficient for each of the independent variables. Sixteen replicated profiles were used in a test-retest measure with the original 16 profiles using Pearson *R* correlations to test the consistency of responses for each respondent.

Results

Table 1 displays the aggregated regression coefficients for each factor, its corresponding Z score, and omega-squared value. At the aggregate level of analysis, the Z scores indicate that: Key success factor stability, timing, lead time, competitive rivalry, educational capability, and industry-related competence are significant in VCs' assessment of probability of survival, i.e., their Z scores exceed 1.645. The sign for the mean regression coefficient for each significant main effect indicates the variable level VCs associated with higher probability of survival, i.e., high key success factor stability (Hypothesis 1), pioneering (H2), long lead time (H3), low competitive rivalry (H4), high educational capability (H5) and high industry-related competence (H6). Therefore Hypotheses 1, 3, 4, 5, and 6 are all supported. Hypothesis 2 was not supported. VCs' significantly used timing of entry in their decision making although in the opposite direction to that hypothesized, i.e., VCs assess a pioneering venture with a higher probability of survival than the probability of survival for a late follower.

On average, the most important criteria for VCs in their assessment of a new venture's probability of

Table 1 Significance and Importance of Decision Criteria

Decision Criteria	β Coefficient	Z Score	Omega Squared
Key Success Factor Stability	0.88	10.52*	0.05
Timing of Entry	0.69	3.57*	0.07
Lead Time	0.66	8.02*	0.03
Competitive Rivalry	0.97	12.14*	0.06
Educational Capability	1.15	14.04*	0.10
Industry Related Competence	2.14	37.77*	0.29
Control Variables			
Scope	0.04	0.18	0.02
Mimicry	0.23	2.46*	0.02

* significant $p < 0.05$

survival is industry related competence ($\omega^2 = 0.29$), educational capability ($\omega^2 = 0.10$), timing ($\omega^2 = 0.07$), competitive rivalry ($\omega^2 = 0.06$), key success factor stability ($\omega^2 = 0.05$), and then lead time ($\omega^2 = 0.03$).⁸ This supports Hypotheses 7(a) and 7(b) (i.e., VCs place higher importance on industry-related competence and educational capability than they do on key success factor stability, timing of entry, lead time, or competitive rivalry). The predictive ability of an individual model with its respective six hold-out profiles was significant for 73% of the VCs ($p < 0.05$), with a mean R^2 of 0.78. The aggregate model also demonstrated predictive ability, significantly predicting the decision policy of 75% of the VCs ($p < 0.05$) with a mean R^2 of 0.78. Ninety-two percent of VCs had significantly reliable responses. Mean test-retest correlation for the sample was 0.69 providing assurances that the new venture decision making task was performed consistently by the participants.

Discussion

This study utilized two perspectives of strategy to understand new venture survival—utilizing both the IO strategy and population ecology literatures provided a deeper understanding of new venture survival than if either literature was used alone. Common to both perspectives is the importance of uncertainty and its relationship with new venture survival (i.e., entrepreneur's uncertainty about the future market and competitors' behavior as well as customers' uncertainty over the newly created industry and the "new" business). These strategy perspectives were used to provide a theoretical framework from which to investigate and interpret VCs' assessments of new venture survival.

This study found considerable consistency between the proposed theoretical framework (based on strategy research) and the decision policies of VCs. Such consistency provided the opportunity to use theory to help us understand which attributes VCs consider and why. For example, VCs are likely to consider the probability of survival higher for those new ventures

that have lower market and industry uncertainty. Furthermore, IO strategy scholars suggest the strategic choices that an entrepreneur makes can impact the survival chances of the firm—the environment will change in unexpected ways and it is important that the new venture be able to recognize those changes and respond appropriately. The need to deal with uncertainty provided insight into why VCs place such high importance on the abilities of the management team.

While VCs' assessment of new venture survival was found to be highly consistent with that proposed by strategy scholars, there was one exception. VCs assess a pioneer as having a higher probability of survival than a late follower, whereas strategy scholars, such as Mitchell (1991), suggest the opposite relationship. It appears that despite the issues of uncertainty and lack of legitimacy facing a pioneer, VCs believe the advantages of being early typically outweigh these initial disadvantages.

Does this represent an opportunity for VCs to improve their assessment of new venture survival (i.e., learn from, and conform with, the relationship between timing and new venture survival proposed by strategy scholars)? Alternatively, is there greater complexity in the relationship between timing and survival that VCs perceive yet is being missed by strategy scholars? For example, maybe the relationship between timing and new venture survival is contingent upon attributes not detailed in the strategy literature (and therefore not included in this study) yet are perceived and used by VCs. This question deserves further consideration by researchers of both strategy and VCs' decision making.

Possible Limitations

Conjoint analysis is used in this study because it accounts for a number of biases and errors that are pervasive in previous research. This study focuses on concurrent, rather than retrospective, techniques of obtaining and analyzing decision making. The technique itself has limitations. Through design of this study and its instrument, these limitations have been addressed (eliminated or minimized).

One such issue is this study's use of hypothetical ventures and environments. However, research into a

⁸ The control variables were relatively unimportant in VCs' assessments of survival.

variety of judgments provides evidence that hypothetical representations are useful for capturing real policies (c.f., Chaput de Saintonge and Hathaway 1981, Riquelme and Rickards 1992). "Professional judgment may therefore involve some abstract coding of the cues, similar to that provided by policy capturing tasks" (Brehmer and Brehmer 1988, p. 89). Use of hypothetical ventures and environments in conjoint analysis is still open to a threat of external validity but is a step towards actual decision making, i.e., more valid inferences can be made using conjoint analysis than those studies based on espoused criteria alone (Stewart 1993).

Sometimes an orthogonal cue set (as used in this study) can lead to the presentation of a profile that simply does not exist in the judge's experience, resulting in possible inconsistency. However, in the pretest all cases were deemed realistic.

There may also be questions regarding the generalizability of the results. The sample included only Australian VCs. Some study participants commented that Australian VCs are similar to those from the United States due to similarities in education,⁹ training, types of investments, as well as the deregulation of the financial industry, which permitted a number of foreign VC firms to enter the Australian industry. However, other study participants commented that the Australian VC industry was relatively new (less than ten years old) and therefore not as mature as the VC industry in the United States or United Kingdom. Therefore, while Australian VCs are likely to be highly similar to those of the United States and United Kingdom, care must be taken in generalizing these results outside Australia.

Conclusion

This study investigated whether VCs' "in use" assessment policies of new venture survival are consistent with those proposed in the strategy literature. The primary finding of this study is that VCs' assessment policies are predominantly consistent with those pro-

posed by strategy scholars. Such consistency provides an opportunity to utilize established theories to help us understand why VCs consider certain criteria in their assessment of new venture survival as well as why some criteria are more important in their assessment than others. VCs consider the level of uncertainty and the ability of the management team to minimize and/or deal with changes in the external environment in their assessment of new venture survival. This approach addressed some of the theoretical limitations of previous research into VCs' decision making.

This study also addressed research method limitations of previous research by (1) acknowledging potential differences in a VCs' decision policy for businesses in different stages of development and (2) investigating the "in-use" decision policies of VCs rather than their "espoused theories of action."

Therefore this study increases our understanding of how VCs assess a new venture in terms of its likely survival. Such an understanding could help entrepreneurs better address their requests for funding to those criteria VCs believe to be most important. Moreover, new ventures that lack a management team with industry-related competence and educational capability would be well advised to acquire these qualities before seeking venture capital or be prepared for the VC to require an appropriate management team be put in place. VCs may use these findings to better understand their own assessment policies, which, in turn, provides the opportunity to increase evaluation efficiency. A better understanding of the "in use" assessment policies of experienced VCs' could also help train inexperienced VCs to make more accurate assessments of new venture proposals.

This study leads to other opportunities for future research, namely, more VC decision making scholars need to utilize theory to drive their research. This is the first study, to the author's knowledge, to use an overarching theory to derive the criteria to be investigated in VCs' assessment of new venture survival. However, we are not proposing that strategy is the only literature that can provide insight into VCs' decision policies; in fact we encourage the investigation of VCs' decisions from a number of different theoretical perspectives. Not only might the use of different theories provide a deeper

⁹ A number of Australian VCs had MBAs from U.S. institutions. Furthermore, Australian business schools are modeled after their U.S. counterparts.

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understanding of VCs' decision policies but these different theories could provide avenues for VCs to improve the accuracy of their assessments. Similarly, using theoretical frameworks to study the "in-use" decision policies of expert VCs might provide important feedback to current theories suggesting the need for theory modifications and/or new theories.

Further research is still required into the "in-use" decision policies of VCs in their assessments of different aspects of performance and for businesses in different stages of development. We know that VCs' overall investment decision can be predicted from their risk and return perceptions, but little empirical research has been done into how VCs combine these two perceptions to form an overall evaluation of investment potential.

This study focuses on VCs' assessment of new ventures, i.e., investment proposals seeking seed to development capital. An interesting question then becomes: Do VCs weight criteria differently in their assessments of businesses in different stages of development? For example, where uncertainties over the future environmental conditions and the appropriateness of specific strategies are reduced (e.g., mezzanine financing or leveraged buyouts), do VCs rely less on the quality of the management team in their assessment of a firm's probability of survival? Do VCs use different criteria for their assessments of businesses in later stages of development? Do VCs weight and/or use different criteria for their assessments of likely profitability or for global measures of success? Much important research remains to be done.

Appendix A

Attributes, Levels, and Definitions

Criteria	Levels	Definition
Key Success Factor Stability	High	requirements necessary for success will not change radically during industry development
	Low	requirements necessary for success will change radically during industry development
Timing of Entry	Pioneer	enters a new industry first
	Late	enters an industry late in the industry's stage of development
	Follower	

Criteria	Levels	Definition
Lead Time	Long	an extended period of monopoly for the first entrant prior to competitors entering the industry
	Short	a minimal period of monopoly for the first entrant prior to competitors entering this industry
Competitive Rivalry	High	intense competition among industry members during industry development
	Low	little competition among industry members during industry development
Educational Capability	High	considerable resources and skills available to overcome market ignorance through education
	Low	few resources or skills available to overcome market ignorance through education
Industry Related Competence	High	venturer has considerable experience and knowledge with the industry being entered on a related industry
	Low	venturer has minimal experience and knowledge with the industry being entered or related industry
<i>Control Variables</i>		
Scope	Broad	a firm that spreads its resources across a wide spectrum of the market, e.g., many segments of the market
	Narrow	a firm that concentrates on intensively exploiting a small segment of the market, e.g., targeting a niche
Entry Wedge Mimicry	High	considerable imitation of the mechanisms used by other firms to enter this, or any other industry, e.g., a franchisee
	Low	minimal imitation of the mechanisms used by other firms to enter this, or any other industry e.g., introducing a new product

Appendix B

CASE 3: Venture AMN

1. This venture's entry wedge mimicry—*low*.
2. This venture's educational capability—*low*.
3. This venture's timing of entry—*late follower*.
4. This venture's market scope—*broad*.
5. This venture's industry related competence—*low*.
6. The industry's first entrant's lead time—*short*.
7. The industry's competitive rivalry—*low*.
8. The industry's key success factor stability—*high*.

SHEPHERD

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Assessment 3: Probability of Survival

Based on the above venture description (using a 10 year time horizon),
how would you rate the probability that this venture will survive?
(Circle the number that best represents your response)

Very Low																			Very High
Probability																			Probability
of Survival	1	2	3	4	5	6	7	8	9	10	11								of Survival

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